

Chapter 26

Production of Powders of Lead, Tin, Bismuth, and Their Alloys

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INTRODUCTION

Lead, tin, and bismuth are distinguished by a low melting point. Their typical physical properties are represented in Table 26.1.

Due to this low melting point, the atomization process is simplified, and there are fewer problems with the selection of the refractory material for manufacturing melting furnaces, fused metal tubes, and atomizers [1–4]. An opportunity for hot gas atomization (above the melt temperature) that increases the stability of the atomization process due to melt drops welding on the atomizer is presented. This allows the use of close-coupled atomizers with the formation of a melt film on the nozzle edge [2–5] (see Chapter 4).

Currently, most lead, tin, and bismuth powders are produced by atomization because it is a convenient and cost-effective technique for the production of large powder batches while ball milling is popular for lead oxide manufacturing and bismuth contains thermoelectric material production.

LEAD POWDER PRODUCTION

Atomization

A typical lead powder production process based on hot gas atomization is described in book [1]. The bars of lead brands (CO and C1 brands in Russian Standard GOST 16138) serve as the raw material. Superheating the molten lead above 820 K noticeably decreases the surface energy and viscosity, resulting in fine powder making. However, the growing sublimation and high toxic lead vapor confine the superheat up to 720 K.

Lead bars are melted in a crucible. The melt is superheated up to 770 K and poured into a tundish, where a constant melt level is maintained. From the tundish, through a calibrated bottom orifice of 4–5 mm in diameter, metal flows into the atomization action zone, where it is atomized by an atomizer with an annular jet. The air blast at 0.6–0.8 MPa range pressure is heated in order to provide the temperature in the atomization zone in the 470–520 K range. To prevent drops sticking to the chamber walls, the vertical metal gas stream is deflected in a horizontal direction by auxiliary airflow. Air consumption for atomization ranges from 1.8 to 2.2 m³ air per 1 kg melt. Powder from the chamber is transferred to sieving and the used air is subjected to dedusting in a cyclone and bag filter.

Table 26.2 gives the chemical and sieve analysis data for several commercially available lead powders produced by atomization. Air-atomized lead particles are roundish in shape. Fig. 26.1 illustrates typical particle shapes and surface topography.

TABLE 26.1 Typical Properties of Lead, Tin, and Bismuth

Property	Unit	Value		
		Lead	Tin	Bismuth
Physical properties				
Density ^a	g/cm ³	11.10	7.30	9.8
Crystal type ^b		fcc	Tetra.	R
Atomic diameter	nm	0.350	0.316	0.326
Melting point	K	601	505	544.2
Boiling point	K	2013	2543	1833
Specific heat	kJ/(kg·K)	130	226	120
Heat of fusion	kJ/mol	5.0	7.2	11.3
Heat of vaporization	kJ/mol	178	290	105
Solid-vapor surface energy	J/m ²	0.61	0.68	...
Elastic modulus at 298 K	GPa	200	55	32
Linear coefficient of thermal expansion from 298 to 373 K	μm/(m·K)	27.56	22.4	13.3
Thermal conductivity at 298 K	W/(m·K)	35.3	66.6	7.8
Electrical resistivity	μΩ·m	0.207	0.115	1.07
Mechanical properties				
Ultimate tensile strength	MPa	12–13	10–50	5–20
Hardness HB	MPa	30–40	50	93
Elongation	%	55	40	...
^a At room temperature.				
^b Crystal type: fcc, face-centered cubic (polymorphic at temperatures below 286.2 K); Tetra, tetragonal; R, rhombohedral.				
From Fedorchenko IM. Encyclopedia of inorganic materials. Kiev: Glavnaja Redakcija USE; 1977 [in Russian], German RM. Sintering theory and practice. New York: John Wiley & Sons Publishers; 1996.				

Ball Milling

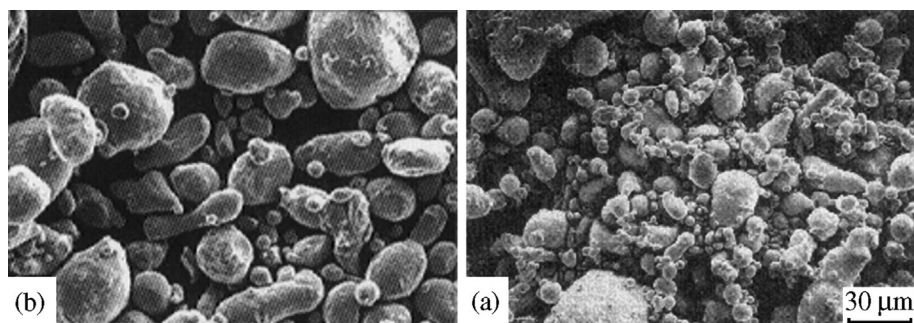
Two main methods, the Barton pot process and the ball mill method, are used for the production of lead oxide, also called lead monoxide, as the key component of automotive lead-acid batteries [6–8]. There are three lead oxides with commercial applications: lead suboxide (2PbO, Pb), also known as gray/lead oxide (battery oxide); litharge (PbO) red lead oxide (Pb₃O₄); or lead ortho-plumbate (PbO₄^{4−}).

The Barton pot process uses a cast iron pot with an upper and lower stirrer rotating at different speeds. Molten lead is fed through a port in the cover into the pot, where it is broken up into droplets by high-speed blades. Heat is supplied initially to develop an operating temperature from 643 to 753 K [7]. The exothermic heat from the resulting oxidation of the droplets is usually sufficient to maintain the desired temperature. The oxidized product is removed from the pot by an air stream. Automatic shaker-type fabric filters, often preceded by cyclone mechanical collectors or settling chambers, are the common choice for collecting lead oxides.

The ball mill method is based on the initiation of lead oxidation in air by heat generated during milling. Lead is fed into the rotating drum in the form of small cylinders. The feed rate is controlled automatically to maintain optimum lead quantities inside the drum. The lead oxide, produced by abrasion between the lead cylinders in a stream of air, is removed and transported to the baghouse by the same airflow. The aspiration system is specifically designed to take out only those particles within the usable size range (fine and very fine). Coarser particles remain in the drum until further oxidation; no screening or return of excessively large particles to the mill is required. The technical data of the SOVEMA “S” Series lead oxide mills [8] are represented in Table 26.3.

TABLE 26.2 Properties of Commercially Available Lead Powders Produced by Atomization

Property	Unit	Standard Grades		
		IIICA ^a	63 ^b	CSC ^c
Nominal composition				
Lead	wt%	99.7 min	99.7 min	99.6 min
Trace elements				
Antimony	ppm	5 max	...	...
Arsenic	ppm	5 max	...	...
Bismuth	ppm	50 max	...	...
Copper	ppm	10 max	...	...
Iron	ppm	20 max	...	...
Oxygen	wt%	0.3 max	...	...
Total impurities	wt%	0.3 max	...	...
Calcined residue after treatment by nitric acid	wt%	0.01 max	...	...
Sieve analysis				
−106 +80 μm	wt%	...	...	5 max
−80 +75 μm	wt%	1.0 max	nil	
−75 +63 μm	wt%	1.5 max	0.2 max	15 max
−63 +45 μm	wt%	...	10 max	
−45 μm	wt%	75–90	...	85 min
Apparent density	g/cm ³	5–6	4.5±0.5	4.5–6.0
^a The requirements of CIS Standard GOST 16138.				
^b Pometon S.p.A. Standard grade.				
^c Chang Sung Corporation.				

**FIG. 26.1** Scanning electron micrographs of atomized (A) tin and (B) lead powders.

Comparison between the reactivity and productivity of the two production processes is characterized as the reactivity of lead oxide. In terms of battery performance, the ball mill method has an undeniable advantage. Therefore, the powder produced by this method is much more active due to the larger specific surface of the powder and also to the absence of orthorhombic (yellow) lead. In terms of productivity and production costs, the Barton pot method has a considerable advantage.

TABLE 26.3 Technical Data of the SOVEMA “S” Series Lead Oxide Mills

Components and Characteristics	Unit	Mill Type		
		S 300 (8T)	S 600 (14T)	S 1000 (24T)
Lead oxide features				
Rate of oxidation	%	60%–80% (metallic lead content 40%–20%)		
Apparent density	g/cm ³	1.4–1.6		
Acid adsorption	mg/g	240–280 mg H ₂ SO ₄ /(g lead oxide)		
Particle size: typical distribution by 70% lead oxide content	µm	50% of the sample < 2.0µm; 75% of the sample below <5.0µm; 90% of the sample below <7.0 µm		
Electric requirements				
Mill				
• Exhaust fan	kW	37	75	132
• Feed conveyer	kW	0.75	0.75	0.75
• Return conveyer	kW	0.55	0.55	1.1
Filter				
• Exhaust fan	kW	13.2	18.4	26
• Cleaning fan	kW	5.5	7.5	9
• Extraction conveyer and valve	kW	0.75	0.75	0.75
• Air flow control shutters	kW	0.15	0.15	0.15
Water requirement				
Flow rate	L/min	30L/min at 0.2MPa		
Average consumption	L/h	300	200	300
Compressed air	MPa	0.6MPa max – 0.4MPa min		
Machine weight (Mill + filter)	kg	8500	15,000	23,000

Electrolysis

Methods of lead powder production by electrolysis render possible the manufacture of high purity powders. Electrolysis process parameters are adduced in [Table 26.4](#).

Ultrafine lead powders, usable in lubricant compositions for formation of a protective layer against radiation, are produced by thermolysis. Alkylamide of oleic acid is mixed with lead formate in proportions ensuring 20% lead content in the mixture, which is then processed at 510 K for 1 h. As a result, powder with roundish particles and sizes ranging from 0.2 to 0.8 μm is produced.

TIN POWDER PRODUCTION

For tin powder production, the horizontal cross-jet air atomization technique is often used [13]. Pressures commonly employed range from 345 to 1725 kPa. Compressed air is preheated in a gas-fired heat exchanger to prevent freezing of the tin within or around the nozzle orifice arising from the chilling effect of the expanding air as it is streaming through the nozzle. The air stream with atomized particles enters the horizontal atomization chamber, where it is sucked by a fan into a cyclone where the powder is collected. The finest particles, which have not separated in the cyclone, are then retained on a filter.

TABLE 26.4 Production of Electrolytic Powders of Bismuth, Tin, Lead, Antimony, and Their Alloys

Method	Electrolyte Composition	Current Density (A/m ²)	Temperature (K)	Electrolysis Conditions and Ready Powder Properties
<i>Bismuth</i> [9]				
EDC	59.5 g/L Bi, 100 g/L NH ₄ Cl; 65.0 g/L glycerin; (pH from 0.2 to 2); 0.2% oleic acid	3.0	278	Cathode rotation 70 rpm; upper layer is mixture of cyclohexane and toluene; particle sizes: 0.05–0.1 μm
EDC	59.5 g/L Bi; 20.0 g/L HCl; 150 g/L NH ₄ Cl; oleic acid 0.5% in toluene	3.0–4.0	288–298	Cathode rotation 60 rpm; ready particulate product is aggregates of small particles.
<i>Tin</i> [10]				
EDC [10]	31.3–62.6 g/L Sn 0.5% Oleic acid	0.5–8.0	278–333	Top layer solvent is toluene; powder particle sizes: 0.2–0.5 μm
CED	12–119 g/L Sn	...	293–333	Substituting metal: Zn; solution: SnCl ₄ ; drum rotation 12 rpm; particle sizes <300 μm; apparent density 1.51 g/cm ³
CED	12–119 g/L Sn	...	300	Substituting metal: Zn; solution: SnCl ₂ ; rate of solution depletion 39.7 g/(L·h); average size of particles 222 μm; apparent density 0.67 g/cm ³
CED	12–119 g/L Sn	...	300	Substituting metal: Al; solution SnCl ₂ ; rate of solution depletion 53 g/(L·h); average size of particles 249 μm; apparent density 0.56 g/cm ³
CED	Sn saturation	—	300	Substituting metal: Zn; solution SnSO ₄ ; rate of solution depletion 10.7 g/(L·h); average size of particles 154 μm ; apparent density 1.08 g/cm ³
<i>Lead</i>				
WS	20–30 g/L Pb; 100–160g/L NaOH; 26.7–40 g/L NaCl; KCl up to 50.0 g/L	0.2–1.5	293	Solution on the basis of PbCl ₂ ; lead anodes; current efficiency: approx. 100%
WS	7.5 g/L Pb; 200 g/L NaCl; 130 g/L NaOH	0.4–1.5	293	Current efficiency: 70–82%; precipitate deposition time is 5 min
WS	40 g/L Pb; 300 g/L NaCl	0.5–2.0	318	Current efficiency: 80.6–89.1%; voltage on the cell: 3–4.7V
WS	28 g/L Pb; 100 g/L NaOH; 29 g/L Na ₂ CO ₃	3.0–4.0	291–296	Precipitate deposition time: 40–60 min
M	0.2–0.9 g/L Pb; NaOH	5.0–12.0	613–643	Lead in the melt is in form of Na ₂ PbO ₂ ; nickel anodes
<i>Lead-Tin</i>				
M	10 g/L Pb; 59.6 g/L SnCl ₂ ; 26.4 g/L KCl	5.0–6.0	510–550	Lead is contained in the melt as PbCl ₂ ; current efficiency: 75%; 93–95% Pb contains in ready product; average size of particles 100 μm
<i>Lead-Bismuth</i> [11]				
EDC	35 g/L (Pb+Bi); 50 g/L NH ₄ NO ₃ ; 30 mL/L HNO ₃	2.0	293	Top layer: 0.03% oleic acid solution in toluene; ratio Pb to Bi is 10 to 1 in ready powder; current efficiency: 45–50%; particle sizes: 0.1–5 μm

Continued

TABLE 26.4 Production of Electrolytic Powders of Bismuth, Tin, Lead, Antimony, and Their Alloys—cont'd

Method	Electrolyte Composition	Current Density (A/m ²)	Temperature (K)	Electrolysis Conditions and Ready Powder Properties
<i>Bismuth-Tin [12]</i>				
EDC	(30–60) g/L (Bi+Sn); 50 g/L (NH ₄)NO ₃ ; 30 mL/L HNO ₃ conc.	2.0	293	Alloys contains 70 to 90% Bi; particle sizes: 0.1–5 μm
EDC	33.2 g/L Bi; 31.4 g/L Sn; 100 g/L NH ₄ Cl; 30 g/L HCl	1.5	298	Cathode rotation 60 rpm; upper layer: toluene with 0.5% oleic acid; current efficiency: 50%; alloy consists about 80% Bi; salts: chlorides
<i>Tin-Antimony</i>				
EDC	(63–200) g/L (Sn+Sb) (80–100) g/L NH ₄ F (pH = 3.54)	0.2–0.6	298	Ratio SnCl ₂ to SbCl ₃ is 1.0 to 1.0; alloys: mixture of tin and intermetallides; current efficiency: 60%–85%; particle sizes: 0.1–2 μm
<i>Technique:</i> WS, water solution electrolysis; EDC, electrolysis from double-layer cell (see section “Reduction from solution through film of surfactants” in Chapter 8); CED, contact electrolytic displacement (see section “Contact displacement method (Cementation)” in Chapter 8); M, electrolysis from the melt [composition—% (by mass)].				

The vertical melt atomization technique also finds application for tin powder production. One such unit is shown in Fig. 26.2 [14]. It consists of a reservoir to which molten metal is fed by gravity. From the reservoir, the metal flows into the tundish. The flow is controlled by an electromagnetic arrangement device so as to maintain a constant level in the tundish and, therefore, a constant flow rate to the atomizer. The reservoir, tundish, batcher, atomizer, and the fused metal tubes connecting them are provided by electric heaters. The atomizer is mounted on the lid of the cylindrical atomization chamber into which a jet of melt is atomized by an annular air jet, forming a metal-air stream. Cold air is fed into that stream with directed jets formed by an exhaust manifold with apertures placed under the chamber lid.

The additional cooling by means of air jets increases the turbulence of the metal-air stream, raises the cooling rate of the particles, and decreases the adhesion of powder on the chamber walls. Atomized powder accumulates in the container at the bottom of the atomization chamber. The finest particles, which have not separated in the chamber, are then retained on a filter.

The closed nozzle design [14] has been used (Fig. 26.3). The atomizer was equipped with changeable parts, nozzles, and sleeve to ensure various operating modes. The nozzles differ by arrangement of the discharges and their diameters: one central discharge for melt out flow (a) and three radial discharges with axes perpendicular to the outside wall (b) of the nozzle. Table 26.5 contains the technological parameters of tin powder production by means of a vertical melt atomization technique.

The physical properties of tin powder depend on atomization conditions and sieving procedures. Coarse grades are atomized by larger metal streams using lower air pressures. The finer grades of powder require higher pressures and a small molten metal stream. Sieving is an independent means of controlling the size of the largest particles in the powder.

Table 26.6 contains data about properties of commercially produced tin powders. Impurities in the tin melt stock carry over to the powder. The tin powder purity is directly related to the purity of the ingot melted.

The production of active powder used as a catalyst is based on the transformation of white tin (β) into gray tin (α) at a temperature below 286.2 K. The gray tin modification has a cubic diamond-type structure with lattice spacing $a = 6.4891 \text{ Å}$. This conversion process takes place at temperatures below 256.4 K with significant volumetric expansion ($\Delta V = 26.6\%$) and depends on temperature and metal purity. Powder particles have an irregular fission-fragment shape with a rather developed surface, which is 30–40 times that of the atomized powder.

By means of electric explosion of tin wire in an ethyl alcohol medium, a powder with particle sizes from 13 to 20 μm is produced. The powder particle sizes were decreased still further two to three times by including 1.0 wt% cadmium or 0.56 wt% bismuth in the melt.

Ultrafine tin powders with particle size ranging from 0.3 to 0.8 μm are produced by electrochemical precipitation from a double layer bath consisting of an aqueous phase represented by chloride/fluoride electrolyte (150 g/L SnCl₂·2H₂O

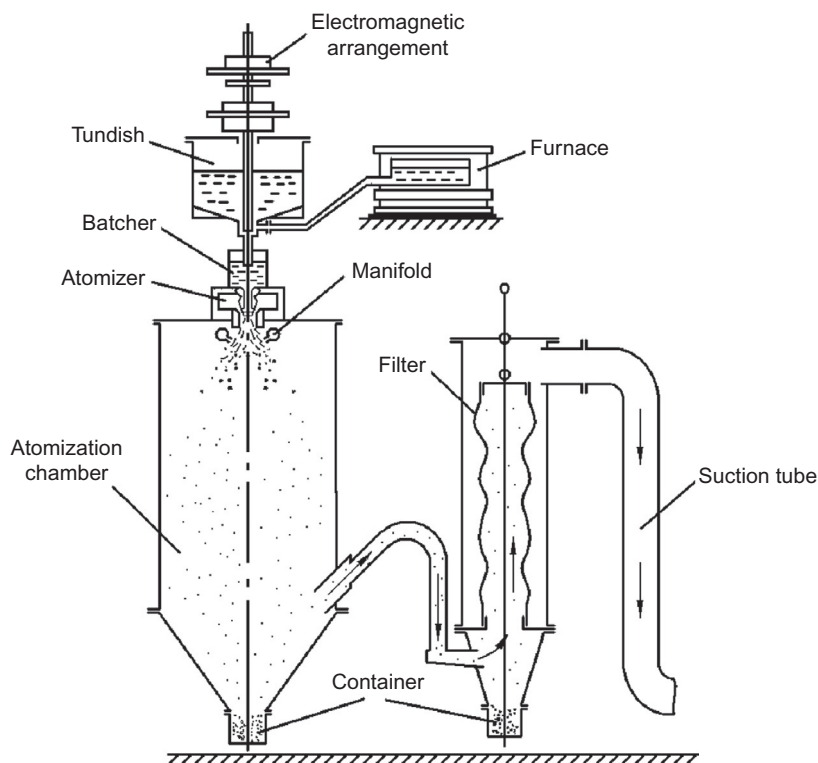


FIG. 26.2 Vertical melt atomization technique of tin powder production.

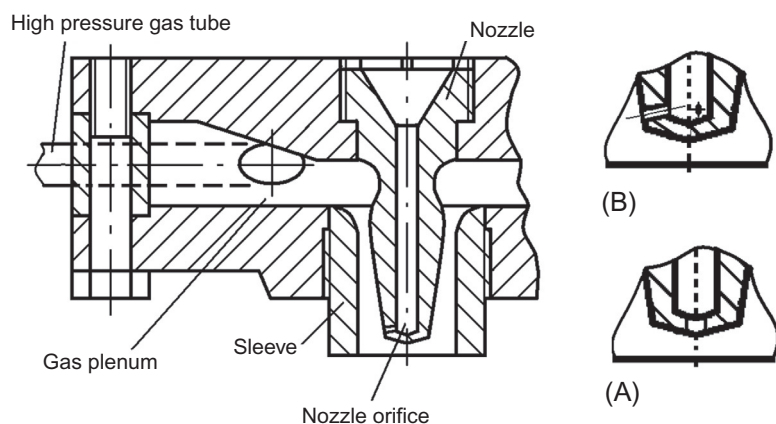


FIG. 26.3 Close-coupled atomizer with changeable nozzles and sleeves: (A) nozzle with three radial discharges with axes perpendicular to the outside nozzle wall; (B) nozzle with one central orifice for melt outflow.

and 60 g/L NH_4F) and an organic medium – a solution of oleic acid in toluene [10]. At current density in the range from 50 to 500 A/m^2 and temperature ranging from 293 to 333 K, tin is reduced from complex anions $[\text{SnClF}_2]^-$ and appears in the form of black powder with particles being distributed uniformly in the organic layer of the bath. Sodium salts are not desirable due to possible formation of sodium oleate. After electrolysis, the powder is extracted from the organic phase, separated from the organic liquid by centrifugation, and washed free from the electrolyte by hot distilled water (360 K) and an aqueous solution of acetone or ethyl alcohol in relation 1.0–1.0. Washed powders are dried under vacuum at room temperature to constant mass. Oxide phases are absent due to adsorption by oleic acid. Current efficiency reaches 100%. The shape of particles is dendritic with short branches of the second order axes (Fig. 26.4). The structure of powders also depends on the cathode material. Comparison of silver, nickel, stainless steel, and aluminum cathodes shows that coarser particles are formed on silver cathodes, which have the most positive potential, while the finest particles are observed on deposition on the aluminum cathode, which has the most negative potential.

TABLE 26.5 Technological Parameters of Tin Powder Production by Means of Vertical Melt Atomization Techniques

Atomizer Type	Melt Mass Flow (kg/h)	Discharge Time of 1.0 kg Melt (s)	Melt Temperature (K)	Operating Gas Pressure in Atomizer (MPa)	Temperature of Operating Gas in Atomizer (K)	Pressure of Cooling Gas (MPa)	Powder Yield (%) ^a
3w/3.0 ^b	498.3	7.2	623	0.55	623	1.8	87.1
3w/2.8	478.8	7.5	623	0.53	593	1.4	93.2
3w/2.0	264.6	13.0	623	0.50	603	3.0	93.9
1/3.0 ^c	361.2	10.0	623	0.51	593	1.5	83.7
1/2.5	297.4	12.1	623	0.50	633	2.5	66.7
1/2.0	198.7	15.1	623	0.52	658	2.5	94.6
1/1.5	185.0	19.5	623	0.50	658	2.2	98.8

^aPercentage of expended raw material for the various nozzle types.

^bFraction numerator signifies "three radial discharges with axes perpendicular to the outside nozzle wall," in denominator is value of the orifice diameters, mm.

^cFraction numerator signifies "the nozzle with one central orifice for melt outflow," in denominator is value of the orifice diameters, mm.

BISMUTH POWDER PRODUCTION

Bismuth powder can be produced by such processes as the mechanical comminution of bulk bismuth, electrolysis of nitrate and sulfate solutions at high current densities, electrolysis of BiCl_3 in methyl alcohol and glacial acetic acid, electrolysis of BiBr_3 in molten AlCl_3 , reduction of aqueous solutions of bismuth salts with various agents, and cathodic reduction of bismuth oxide.

Electrolytic bismuth powder is produced by the same method as tin powder in a two-layer bath with rotating cathode and soluble anode [9]; electrolyte composition: aqueous phase of 59.5 g/L Bi, 20 g/L HCl and 150 g/L NH_4Cl ; organic phase toluene, and 5% oleic acid. The process is carried out at current density 3–4 kA/m² and at temperature ranging from 288 to 298 K; current efficiency is between 65% and 70%. Ultrafine powder modified by oleic acid is separated from the organic phase by centrifugation, then freed from the electrolyte by hot distilled water (360 K) and an aqueous solution of acetone or ethyl alcohol 1.0–1.0. Powders are dried under vacuum at room temperature to constant mass. Powders produced have particle sizes in the range from 0.1 to 0.5 μm . Other electrolytic production methods for bismuth, tin, and lead powders as well as their alloys [9–12] are given in Table 26.4.

A reduction method of bismuth powder production was described by the authors [15]. Bismuth powder was produced from freshly prepared $\alpha\text{-Bi}_2\text{O}_3$ by reduction with a mixture of molecular and atomic hydrogen at 543 K. The reduction process $\alpha\text{-Bi}_2\text{O}_3 \rightarrow \text{Bi}$ makes possible the synthesis of bismuth powder without any intermediate phases. Up to 573 K, the reduction occurs due to a diffusion mechanism while at higher temperatures, it occurs through a diffusion-kinetic mechanism, resulting in increasing the reaction rate up to six times.

American Elements Company [16] specializes in producing high-purity bismuth powder (up to 99.9999 of BI-M-06-P brand) for use in preparation of pressed and bonded sputtering targets as well as in chemical vapor deposition (CVD) and physical vapor deposition (PVD) processes by thermal and electron beam evaporation, low temperature organic evaporation, atomic layer deposition, and metallic-organic and chemical vapor deposition. Commercially available powder comes in the following particle sizes: <44 μm , <150 μm , 10–50 μm , and submicron (<1 μm).

HEALTH AND ENVIRONMENT PROTECTION

Lead and the manufacture of its inorganic compounds carry grave dangers and harmful conditions. Lead is a highly toxic agent. According to US New Jersey Department of Health and Senior Services, the OSHA (Occupational Safety and Health Administration) legal airborne permissible exposure limit is 0.05 mg/m³ averaged over 8-hour workshift. The occupational exposure limits (OEL) in the workplace atmosphere under legislation in the CIS Standard [17] amounts to 0.005 mg/m³ as

TABLE 26.6 Properties of Commercial Tin Powders

Properties	Unit	Standard Grades									
		ASTM Grade, sample			GOST 9723 ^a			MMP ^b		Pom ^c	Bel ^d
		A	B	C	PO1	PO2	PO3	−100	−600	106	M Tin
Nominal composition											
Tin	wt%	99.8 min	99.8 min	99.8 min	99.1 min	99.0 min	99.0 min	99.85 min	99.85 min	99.7 min	99.85 min
Trace elements, max											
Antimony	wt%	0.04	0.04	0.04	0.05	0.05	0.05	...	...	..	0.04
Arsenic	wt%	0.05	0.05	0.05	0.015	0.015	0.015	...	...	..	0.05
Bismuth	wt%	0.015	0.015	0.015	0.05	0.05	0.05	...	...	...	0.03
Cadmium	...	...	...	...	...	...	...	...	...	...	0.001
Copper	wt%	0.04	0.04	0.04	0.03	0.03	0.03	...	...	...	0.04
Iron	wt%	0.015	0.015	0.015	0.02	0.02	0.02	...	...	...	0.01
Lead	wt%	0.05	0.05	0.05	0.15	0.25	0.215	...	...	...	0.05
Nickel	...	...	...	...	...	...	...	...	...	...	0.01
Oxygen	wt%	...	...	...	0.5	0.5	...	0.15	0.15	...	...
Sulfur	wt%	0.1	0.1	0.1	0.016	0.016	0.016	...	...	...	0.01
Zinc	...	...	...	...	...	...	...	...	...	...	0.005
Sieve analysis											
+150 μm	wt%	48.0	0.3 max	...	...	...	...	0,5 max	...	...	1.0
+125 μm	wt%	...	...	...	...	...	...	...	...	nil	...
+100 μm	wt%	46.5	3 max	...	...	...	...	15. max	...	0,1 max	...
+80 μm	wt%	...	...	...	0.5 max	0.5 max	...	...	...	...	...
+75 μm	wt%	4.0	3–8	...	...	...	...	15–35	nil	...	3.0
+71 μm	wt%	...	...	...	2.0 max	2.0 max	...	...	...	...	...
−63 μm	wt%	...	...	...	...	...	100	...	...	...	...
+53 μm	wt%	...	...	...	...	...	...	10–30	0.5 max	...	...
+45 μm	wt%	1.0	12–25	...	...	...	...	10 max	2.0 max	15–45	...

Continued

TABLE 26.6 Properties of Commercial Tin Powders—cont'd

Properties	Unit	Standard Grades									
		ASTM Grade, sample			GOST 9723			MMP		Pom	Bel
		A	B	C	PO1	PO2	PO3	−100	−600	106	M Tin
−45 μm	wt%	0.5	70–85	98 min	85–95	80 min	95 min	35–55	98 min	...	96.0
Fisher subsieve size, average size of particles	μm	...	10–15	1–3	...	...	...	...	...	...	...
Apparent density	g/cm ³	3.35	3.7–4.2	1.3–2.0	3.0–4.0	3.2–4.2	3.2–4.2	...	...	3.8±0.4	3.3

^aRussian Federation Standard.^bMMP—Makin Metal Powders Ltd (GB).^cPometon. S.p. A., *Metal powders and granules (Italy)*.^dBel—Belmont metals Inc. (USA).

ASTM—American Society for Testing and Materials.

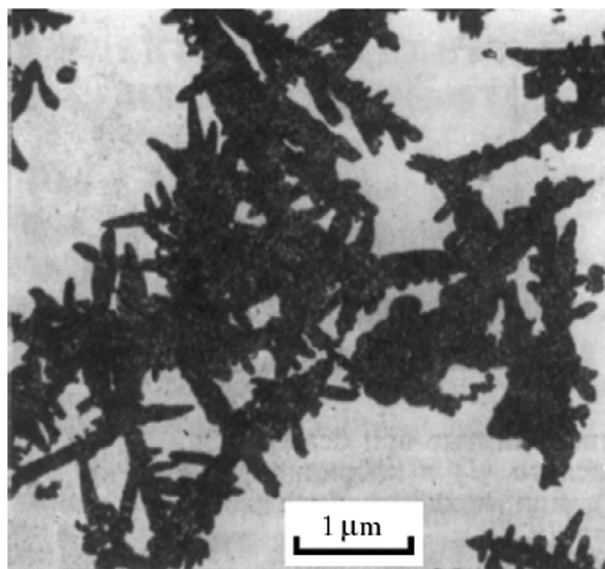


FIG. 26.4 Typical scanning electron micrograph of powder produced by electrochemical precipitation from a double-layer bath.

the shift time weighted average concentration and 0.01 mg/m^3 as peak concentration. It is classed with the extremely dangerous powders, the first class according to the above state Standard.

Because of the high toxicity of lead and its inorganic compounds, a complex of the industrial safety measures of lead powder manufacture is provided. These industrial safety measures include maximum encapsulation of all units and inter-operational transport, maximum mechanization and automation of the processes, remote control of units, and conducting manual operations by means of manipulators in airtight boxes. In vented chambers where equipment is located, an under-pressure is maintained in order to prevent an escape of airborne lead. The lead-containing suction gases are subjected to high-performance cleaning. Lead trace concentrations are continuously controlled in the workplace atmosphere and in the ambient factory air.

The admissible concentration limit of lead in an aqueous medium per toxicological indication is 0.03 mg/L . It is classified as a highly dangerous agent in the second dangerous class according to SanPiN 4630-88 Standard [18].

The occupational exposure limit in the workplace atmosphere of bismuth and its inorganic compounds is 0.5 mg/m^3 . It is placed in the second class of danger, according to the RF Standard [17].

The admissible concentration limit of bismuth in an aqueous medium per toxicological indication is 0.1 mg/L (for the inorganic compounds, including transition elements, subject to total contents of all forms). It is classified as a highly dangerous agent in the second dangerous class, according to SanPiN 4630-88 Standard [18].

The vapors of toluene used for the electrochemical precipitation of tin powder in a double-layer bath process are classed as a third-class danger. Its occupational exposure limit in the workplace atmosphere amounts to 50 mg/m^3 . Toluene is a substance with pencil-beam action demanding the automatic monitoring of its content in air. It is capable of causing allergic illnesses in working conditions. The admissible concentration limit of toluene in an aqueous medium per organoleptic indication, changing the smell, is 0.5 mg/L . It is classed as a moderately dangerous agent in the fourth dangerous class.

Metallic tin in particulate form is a combustible material while solid metal is noncombustible. The self-ignition temperatures of tin powder, containing 98 wt% with particles below $74 \mu\text{m}$ in size in aerogel and suspended solid conditions, are 793 and 933 K, respectively. The low concentration ignition limit (LCIL) is 220 g/m^3 ; the maximum explosion-proof oxygen content in airborne powder (LOC) diluted by carbon dioxide is 15 vol%. The combustibility and explosiveness of tin alloy with 2 wt% lead is characterized by the following data. Self-ignition temperatures of the powder with particles below $74 \mu\text{m}$ in size in aerogel and suspended solid conditions are 703 and 903 K, respectively [19]; the LCIL is 190 g/m^3 ; the maximum explosion pressure by airborne powder content 1000 g/m^3 is 340 kPa; the maximum rate of pressure rise is 12 MPa/s ; the minimum airborne powder ignition energy is 80 mJ; and the LOC diluted by carbon dioxide is 16 vol%.

Products with such characteristics of inflammability and explosion risk are placed in the “moderate explosion hazard” class of danger, according to the Guide to Legislation and Health and Safety in the European powder-making industry [20].

To extinguish burning tin powder, it is necessary to use a mix of potassium and sodium chloride powder (with consumption about 15–50 kg per one square meter), or dry sand. For the voluminal quenching method, argon is used, consuming about 12.0 kg/m³ in the burning zone.

Lead in particulate form is difficult to ignite. Self-ignition temperatures of lead powder with particles below 53 μm in size in aerosol and suspended solid conditions are 473 and 983 K, respectively.

Oleic acid, used for electrochemical precipitation of tin, bismuth, and their alloy powders in the double-layer bath process, is a combustible liquid. Its properties are characterized by the following data. The melting point is 287 K, the boiling point is 653 K, and the heat of combustion 10,370 kJ/mol. Oleic acid is insoluble in water. Its self-ignition and ignition temperatures are 553 and 453 K, respectively. To extinguish burning oleic acid, it is necessary to use bicarbonate of soda powder or the voluminal quenching method by means of carbon dioxide, consumption of about 0.7 kg/m³ in the burning zone.

The cyclohexane, which is also used in the electrochemical precipitation of tin, bismuth, and their alloy powders in the double-layer bath process, is a highly inflammable liquid. Its properties are characterized by the following data. The melting point is 279.5 K, the boiling point is 353.7 K, and the heat of combustion is 3689 kJ/mol. Cyclohexane is insoluble in water. Its self-ignition temperature is 532 K. The lower and upper flame propagation limits in air are 1.3 and 7.8 vol%, respectively. The maximum explosion pressure is 858 kPa, the maximum rate of the flame propagation is 0.436 m/s, the minimum ignition energy is 0.22 mJ, and the burning temperature amounts to 2270 K. The means of extinguishing burning cyclohexane are the same as for oleic acid [19].

Techniques for the prevention of exposure to hazards are described in [Chapter 27](#).

APPLICATIONS

The largest commercial application of lead and lead oxide, key components of lead-acid batteries, is the automotive industry. According to the “Global Automotive Battery Market—Analysis of Growth, Trends, Progress, and Challenges (2015–2020)” report, issued in August 2016, the automotive battery market is expected to grow at a CAGR of 5.29% to a market size of USD XX billion by the year 2020. Automobiles have a large number of functions these days. The new features have put great pressure on car batteries, which have to take on higher loads. The new fuel-saving technologies such as microhybrid systems are also leading to higher battery usage.

Tin powders are used extensively in the production of porous, self-lubricating bronze bearings and as constituents in soldering and brazing powders and pastes. Tin powders are also used in powder metallurgy structural parts, friction disks, clutches, brake linings, bonded resin friction materials, metal graphite brushes, diamond abrasive grinding wheels, bronze filters, peen plating, plasma arc spraying, chemical formulations, additives for rubber and plastics, chemical manufacturing, smokeless powder for pyrotechnics, foams, and tin flake.

The requirements for powders utilized for soldering paste production are a high content of fine particles and a low oxygen content. Some typical solder alloy powder grades used in soldering pastes and creams are listed in [Table 26.7](#).

Because copper and lead have limited solubility in each other, they are difficult to alloy by conventional ingot metallurgy. Copper-lead powder mixtures have excellent cold-pressing properties. They can be compacted at pressures as low as 76 MPa to densities as high as 80% of theoretical density. After sintering, they can be repressed at pressures as low as 152 MPa to become practically nonporous bearings.

Cu-Pb-Sn PM materials are also used to replace solid bronze bearings. They are produced by spreading the powder in a predetermined thickness on a steel strip, sintering, rolling to theoretical density, resintering, and annealing. The end product has a residual porosity of about 0.25%. These materials include Cu-25Pb-0.5Sn, Cu-25Pb-3.5Sn, Cu-10Pb-0.5Sn, and Cu-50Pb-0.5Sn.

Tin, lead, and bismuth powders are used as compacted metal powders in the production of metallic foams. The metal powders are mixed with foaming agents to give a completely homogeneous mixture. The result of the subsequent densification step is a foamable material that, upon heating to temperatures within the range of the melting point, expands into a highly porous cellular solid with a closed pore structure. The typical structure of lead foam is shown in [Fig. 26.5](#). More information about metallic foam technology can be found in [Chapter 11](#).

As is known, the conversion of thermal energy to electrical energy is named as a thermoelectric (TE) conversion [21]. The TE effect can be used for both power generation and electronic refrigeration. Thermoelectric technology and solid-state devices based on the TE effect have a number of advantages, including having no moving parts as well as being reliable and scalable. The technology has therefore aroused worldwide interest in many fields, including waste heat recovery and solar heat utilization (power generation mode) as well as temperature-controlled seats, portable picnic coolers, and thermal management in microprocessors (active refrigeration mode).

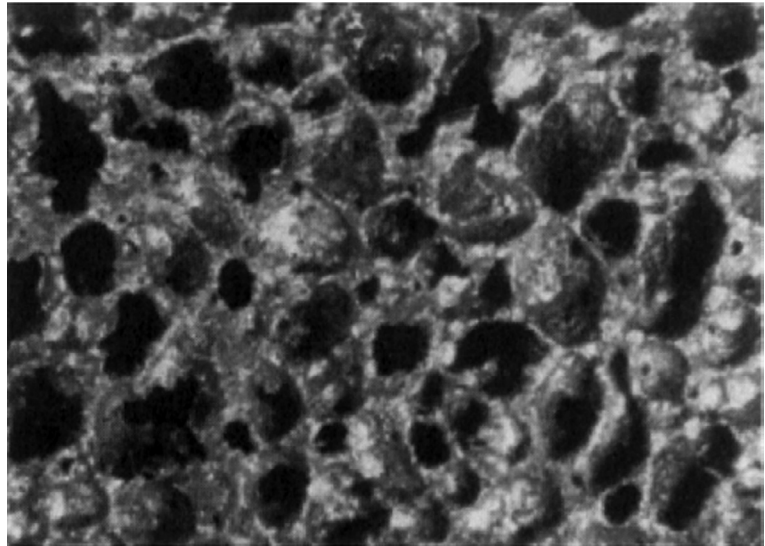
TABLE 26.7 Typical Solder Alloy Powder Grades

Property	Unit	Standard Grades													
		ASTM Grade, Samples						GOST 1499, Samples ^a						Nippon AMP	
		10 B	30 B	50 B	63 B	62 P	96.5TS	ΠOC 10 ^b	ΠOC 40 ^b	ΠOC 61 ^b	ΠOC 90 ^b	ΠOCCy 61-0.5 ^c	ΠOCCy 95-5 ^d	80-20	60-40
Composition															
Main components															
Tin	wt%	8–12	28–32	48–52	61–65	61.5–67.5	95–96	9–10	39–41	60–62	89–91	60–62	94–96	80	60
Antimony	wt%	0.2–0.5	0.2–0.5	0.2–0.5	0.2–0.5	0.3–0.5	0.2–0.5	...	...	...	...	0.2–0.5	4.0–5.0	...	...
Lead	wt%	bal	bal	bal	bal	bal	bal	bal	bal	bal	bal	bal	bal	20	40
Trace elements, max															
Aluminum	wt%	0.005	0.005	0.005	0.005	0.005	0.005	0.002	0.002	0.002	0.002	0.002	0.002	...	...
Antimony	wt%	...	...	...	...	...	...	.05	.05	.05	.05	...	...	...	...
Arsenic	wt%	0.02	0.02	0.02	0.02	0.02	0.02	0.03	0.03	0.03	0.02	0.03	0.05	...	...
Bismuth	wt%	0.25	0.25	0.25	0.25	0.25	0.25	0.1	0.1	0.1	0.1	0.1	0.2	...	...
Copper	wt%	0.08	0.08	0.08	0.08	0.08	0.08	0.05	0.05	0.05	0.05	0.08	0.1	...	...
Iron	wt%	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	0.02	...	...
Lead	wt%	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Nickel	wt%	...	...	...	...	...	...	0.02	0.02	0.02	0.02	0.02	0.08	...	...
Silver	wt%	...	...	...	...	1.75–2.25	3.3–3.7	...	...	...	...	...	...	...	...
Sulfur	wt%	...	...	...	...	...	...	0.02	0.02	0.02	0.02	0.02	0.02	...	...
Zinc	wt%	...	...	...	...	...	...	0.002	0.002	0.002	0.002	0.002	0.002	...	...
Particle sizes	μm	...	...	...	...	...	...	≤50	≤50	≤50	≤50	≤50	≤50	5 ^e	10 ^e
Solidus temperature	K	541	456	456	456	452	494	...		...	...	...	...	...	...
Liquidus temperature	K	572	528	489	456	452	494	...		...	...	...	...	...	...
Soldering range	K	602–642	558–598	486–555	485–526	482–531	524–564	...		...	...	...	...	...	...

^aCIS Standard.^bAntimonousless.^cLow- antimonous.^dAntimonous.^eMedian diameter.

Nippon AMP: Nippon Atomized Metal Powders CO.

FIG. 26.5 Optical micrograph of cross-section of lead foam. From Banhart J, Baumeister J, Weber M. *Metallic foam*. In: *ASM handbook*, vol. 7. ASM International Publishers; 1998. p. 1043–7.



Li et al. [22] focused on thermoelectric conversion of Bi-Te alloys, skutterudite compounds, and Ag-Pb-Sb-Te quaternary systems. A novel method for rapid preparation of Bi_2Te_3 nanosized powders with an average particle size of about 70 nm was developed by Chen et al. [23]. A starting Bi_2Te_3 particulate material of ~ 5 mm average size was subjected by cryogenic grinding in the liquid nitrogen. For comparison, the conventional high-energy ball milling was used to prepare the Bi_2Te_3 nanosized powders. It was found that Bi_2Te_3 does not transform into a nonequilibrium amorphous phase or decomposes during the cryogenic grinding process. Spark plasma sintering was used for powder consolidation. The authors reported that spark plasma sintering showed that cryogenic grinding can produce much finer and good sinterability Bi_2Te_3 powders in comparison with high-energy ball milling.

REFERENCES

- [1] Naboychenko SS, editor. *Handbook of nonferrous metal powders*. Moscow: Metallurgia Publishers; 1997 [in Russian].
- [2] Strauss JT. Close-coupled gas atomization using elevated temperature gas. In: *Proceedings of 1999 powder metallurgy international conference*, vol. 1, San Diego, CA: Metal Powder Industries Federation; 1999. p. 22–33.
- [3] Wolf G, Till K, Scholz R. Hot gas atomisation. In: *Proceedings of 2002 powder metallurgy world congress* Princeton: MPIF Publishers; 2002. p. 3-163–3-177.
- [4] Dunkley JJ, Aderhold D. The fourth dimension in gas atomisation and the influence of gas density on the atomisation of melts. In: *Proceedings of the 4th International Conference on Spray Deposition and Melt Atomization*, SDMA2009 Bremen: Universität Bremen; 2009. p. 1–8.
- [5] Rieken JR, Heidloff AJ, Anderson IE, Mullis A, Chernenkoff RA, Brian W. Development of a close-coupled gas atomization process for fine metal powder production. In: *Proceedings of 2014 Powder Metallurgy World Congress* Orlando, FL: Metal Powder Industries Federation; 2014. p. 02-54–60.
- [6] Pavlov D. *Lead-acid batteries*. Amsterdam: Elsevier; 2011.
- [7] Gravita India Ltd. <http://www.gravitagroup.com/Lead-Oxide-Manufacturing.asp>.
- [8] SOVEMA “S” Series lead oxide mills, di Verona (Italy). [//sovema_lead_oxide_mill.pdf](http://sovema_lead_oxide_mill.pdf).
- [9] Zhelibo EP, Grechanjuk VG. On bismuth fine powders deposition on passivated electrodes. *Sov Powder Metall Met Ceram* 1978;6:1–6.
- [10] Zhelibo EP, Grechanjuk VG. Electrodeposition of tin fine powders from chloride-fluoride electrolytes. *Sov Powder Metall Met Ceram* 1978;7:4–7.
- [11] Fil’ TN, Khimchenko EI, Isaj VN, Romanova AV. Electrodeposition and X-ray diffraction examination of high-disperse lead–bismuth alloy. *Sov Powder Metall Met Ceram* 1977;3:12–9.
- [12] Zhelibo EP, Grechanjuk VG. Effect of electrolyte composition on conditions of tin–bismuth highdispersed alloy formation. *Sov Powder Metall Met Ceram* 1978;8:5–9.
- [13] Ullrich WJ. Production of tin powders. In: *ASM handbook*. vol. 7. ASM International Publishers; 1998. p. 146–7.
- [14] Adamchik LV, Bondarenko SG, Koval’chuk VG, Pavlenko DV, Statsenko ID. Gas-sprayed tin powders. *Sov Powder Metall Met Ceram* 2003;42 (1–2):100–4.
- [15] Chernogorenko VB, Lynchak KA. Production of bismuth powder by the reduction of bismuth oxide with a mixture of molecular and atomic hydrogen. *Sov Powder Metall Met Ceram* 1973;12(5):360–2.
- [16] American Elements Company. <https://www.americanelements.com/bismuth-particles-7440-69-9> [Accessed 31 January 2018].
- [17] CIS Standard GOST 12.1.005-88. Moscow: Publishers of Standards; 1988 [in Russian].
- [18] Sanitary Regulation and Standards of surface waters from pollution. SanPiN 4630-88. Moscow: MinZdrav; 1988 [in Russian].

- [19] Baratov AN, Korolychenko AJ, editors. Handbook Fire- and explosion-proof of matters and materials and means for their quenching. Chimia: Moscow; 1990 [in Russian].
- [20] Guide to Legislation and 'Health and Safety' in the European PM Industry. EPMA; 1997.
- [21] Bechnia K. Fundamentals of Thermoelectrocity. Oxford: University Press; 2015.
- [22] Li JF, et al. High-performance nanostructured thermoelectric materials. NPG Asia Mater 2010;2:152–8.
- [23] Chen X, et al. Preparation of nano-sized Bi₂Te₃ thermoelectric material powders by cryogenic grinding. Prog Nat Sci Mater Int 2012;22(3):201–20.

FURTHER READING

- [24] Fedorchenko IM, editor. Encyclopedia of inorganic materials. Kiev: Glavnaja Redakcija USE; 1977 [in Russian].
- [25] German RM. Sintering theory and practice. New York: John Wiley & Sons Publishers; 1996.
- [26] Lawley A. Atomization. Metal Powder Industries Federation Publishers: Princeton; 2003.
- [27] Banhart J, Baumeister J, Weber M. Metallic foam. In: ASM handbook. vol. 7. ASM International Publishers; 1998. p. 1043–7.